Core Pure 2 · Chapter 4

Volumes of Revolution

Cartesian curves · Parametric curves · Mathematical models

Chapter Overview

Choose the axis

About the x-axis: integrate πy^2 with respect to x.

About the y-axis: integrate πx^2 with respect to y.

Choose the method

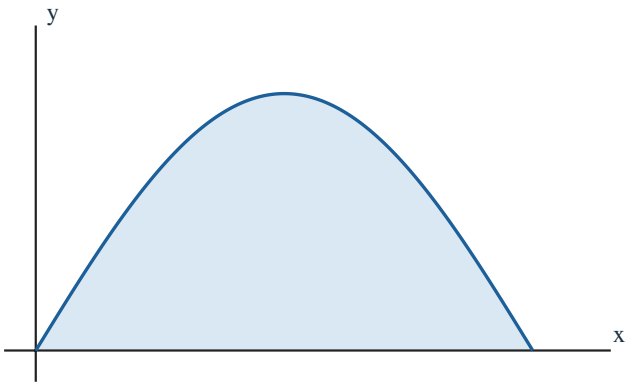
Convert limits when using a parameter.

Subtract inner volumes for hollow regions.

Apply cubic scale factors to volumes in models.

Volumes of Revolution with harder integration 3

The region under $y = \sin 2x$, from $x = 0$ to $x = \pi/2$, is rotated about the x-axis. Find its volume.



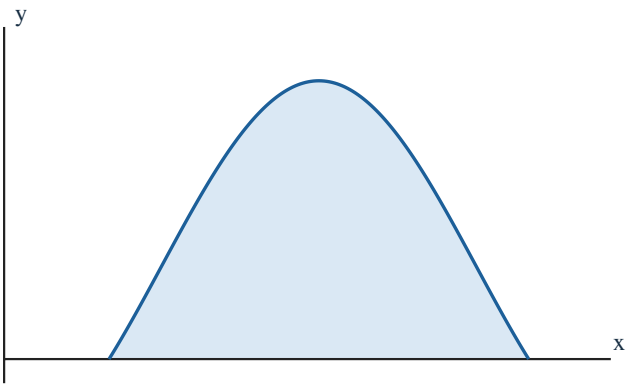
Disc method

Evaluate

Test Your Understanding 4

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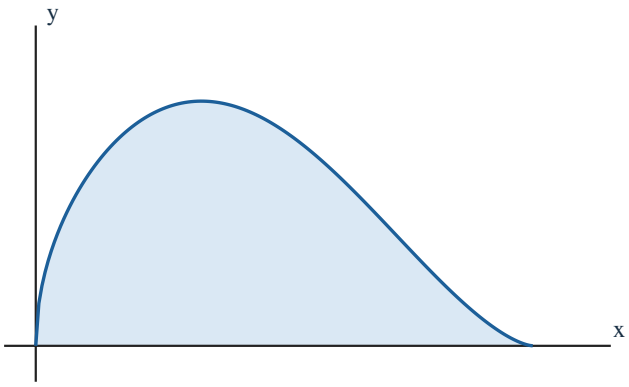
The region between $y = 1 - 2\cos x$ and the x-axis, from its first to second positive roots, is rotated about the x-axis. Find the exact volume.



Limits

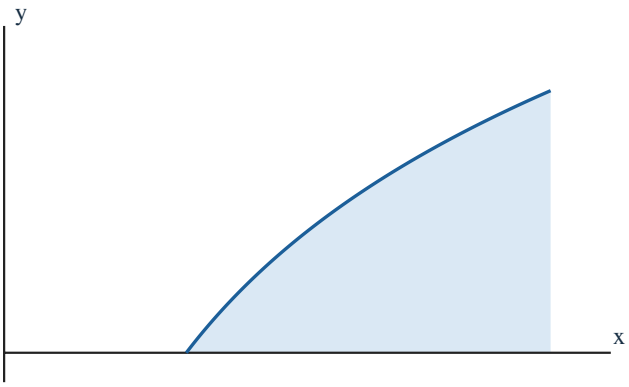
Volume

2. Rotate the region under $y = \cos x\sqrt{\sin 2x}$, $0 \leq x \leq \pi/2$, about the x-axis. Find its volume.



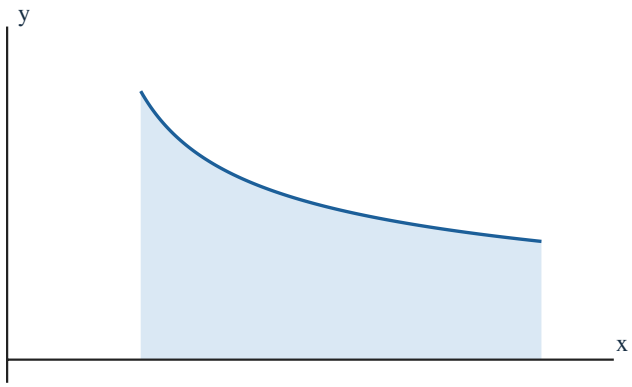
2

3. The region bounded by $y = \ln x$, the x-axis and $x = 3$ is rotated about the x-axis. Find its exact volume.



3

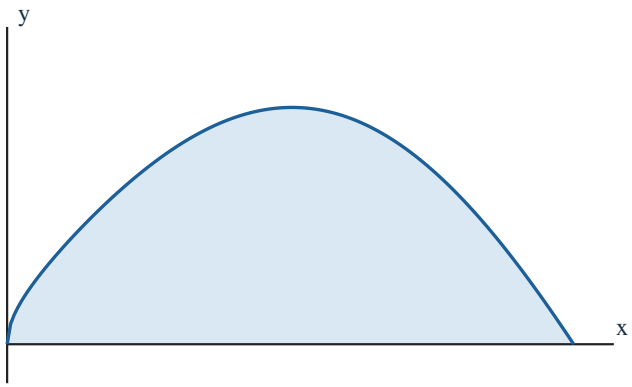
5. A curve satisfies $y^2 = (4x + 3)/((x + 2)(2x - 1))$. Find the volume generated by rotating the region under its positive branch, $1 \leq x \leq 4$, about the x-axis.



Split

Volume

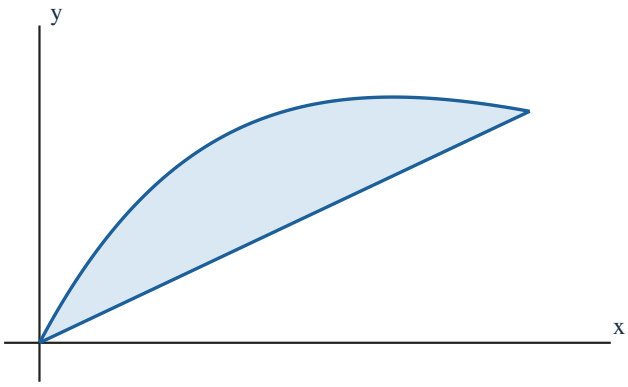
6. The curve $2y^2 = x \sin x + x$ forms a loop from the origin to its first positive x-axis intersection. Find the volume formed by rotating the loop about the x-axis.



Intersection

Volume

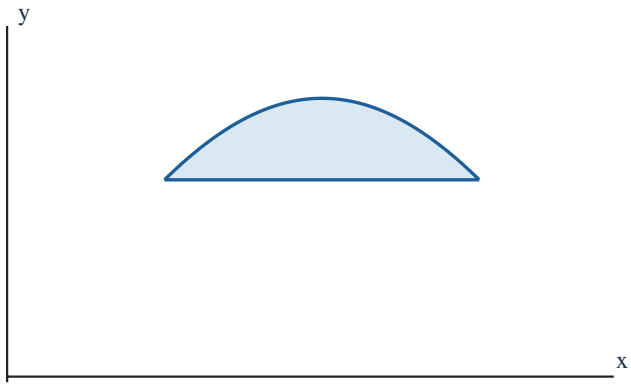
8. The positive region between $y = xe^{-x}$ and $y = x/4$ is rotated about the x-axis. Find its volume to 3 significant figures.



Intersections

Washer method

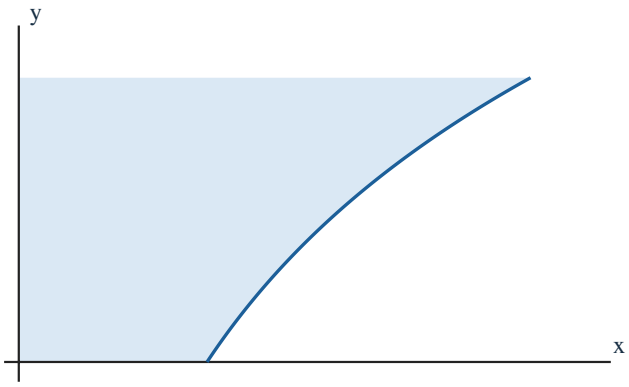
Challenge. Rotate the region between $y = \sin x$ and $y = 1/\sqrt{2}$, for $\pi/4 \leq x \leq 3\pi/4$, about the line $y = 1/\sqrt{2}$. Find the exact volume.



Shift the axis

Revolving around the y-axis 11

The region between the y-axis and $y = 4\ln x - 1$, for $0 \leq y \leq 4$, rotates about the y-axis. Find its volume.

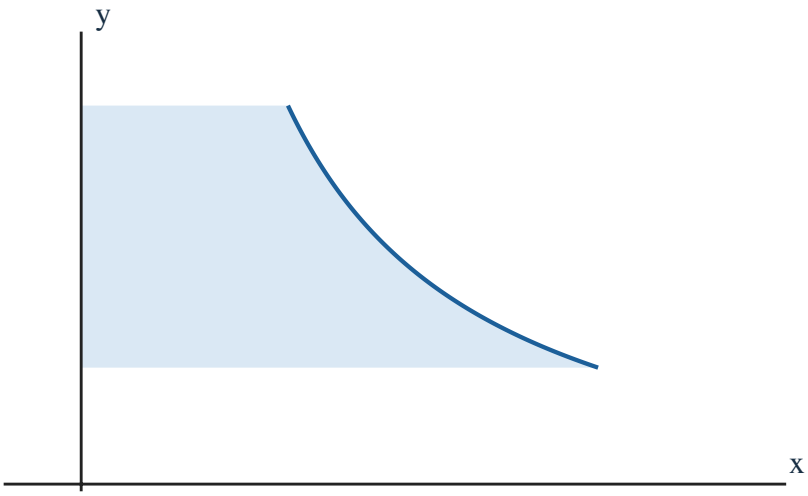


Formula

Evaluate

Exercise 4B - Core Pure Year 2 Pages 81-83 12

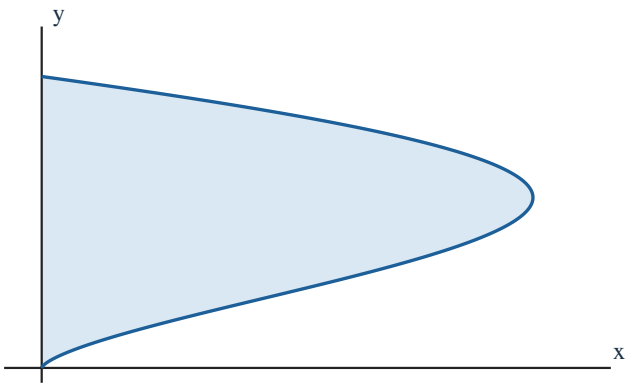
3. The region between $x = 1/(2y + 1)$, the y-axis and $y = 1, b$, where $b > 1$, rotates about the y -axis. Its volume is $\pi/10$. Find b.



Integral

Solve

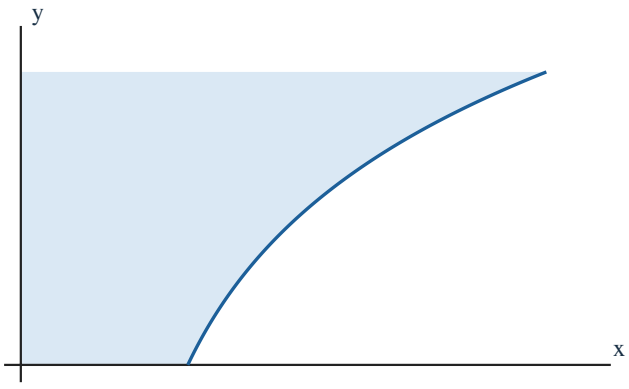
4. The first lobe of $x = \sqrt{y}\sin y$, $y \geq 0$, is bounded by the curve and the y-axis. Find its upper endpoint and the volume generated about the y-axis.



Endpoint

Volume

5. The region between $y = 3\ln(x-1)$, the y -axis, $y = 0$ and $y = 5$ is rotated about the y-axis. Find its exact volume.

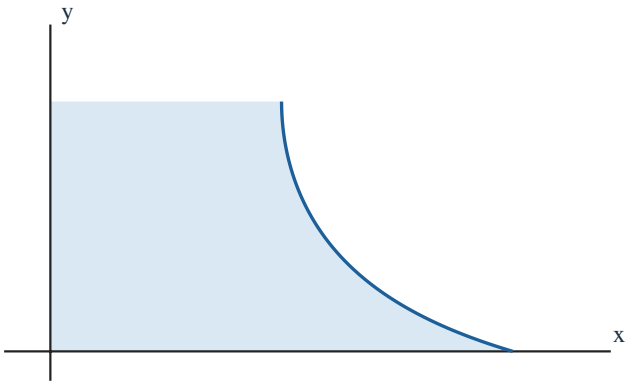


Rearrange

Volume

6a. Write $\cos y + \sqrt{3}\sin y$ as $R\cos(y - \alpha)$.

b. Find the volume about the y-axis of the region between that axis and $x = 1/(\cos y + \sqrt{3}\sin y)$, for $0 \leq y \leq \pi/3$.

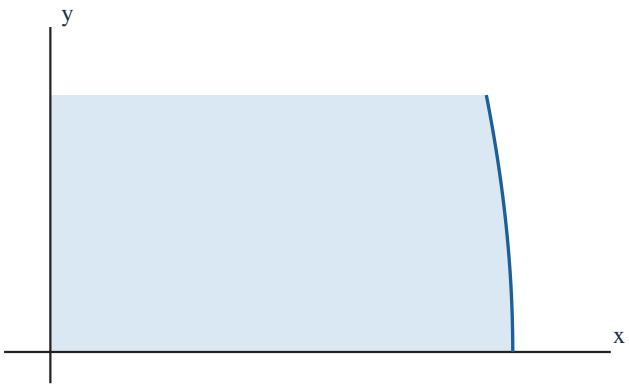


a

b

7a. Use $u = 2^y$ to evaluate $\int_0^1 2^y/(2^y + 1)^2 dy$.

b. Hence find the volume generated about the y-axis by $x = \sqrt{2^y}/(2^y + 1)$, between $y = 0$ and 1.

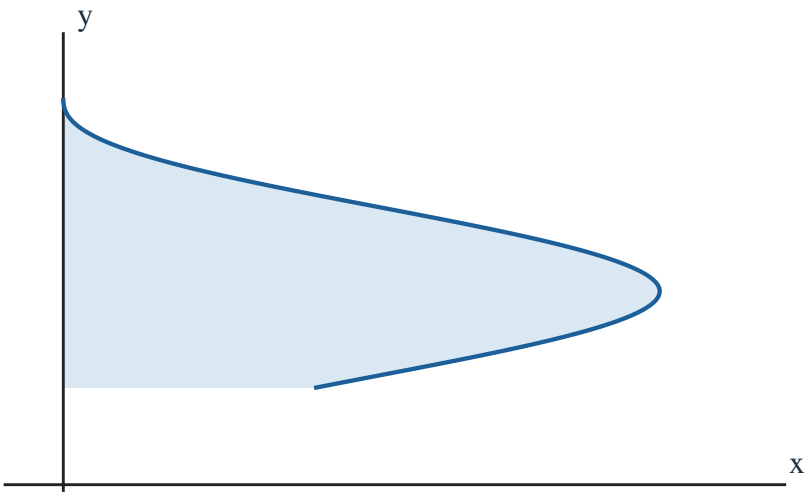


a

b

8a. Prove $\sin^5\theta = (10\sin\theta - 5\sin 3\theta + \sin 5\theta)/16$.

b. Rotate the region between $x = \sin^2 y\sqrt{\sin y}$ and the y-axis, $\pi/4 \le y \le \pi$. Find the exact volume.



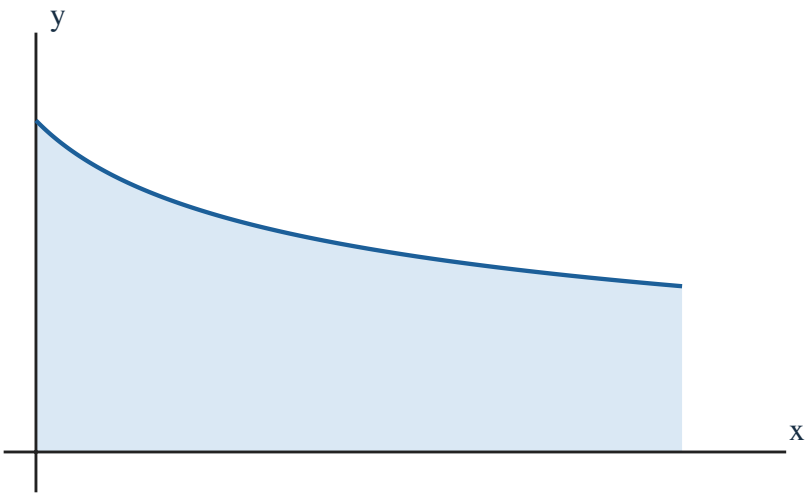
a

b

Volumes of revolution for parametric curves 18

Parametric curves

$x = t(1+t),\ y = 1/(1+t),\ t \ge 0$. Rotate the region under the curve from $x = 0$ to 2 about the x-axis.

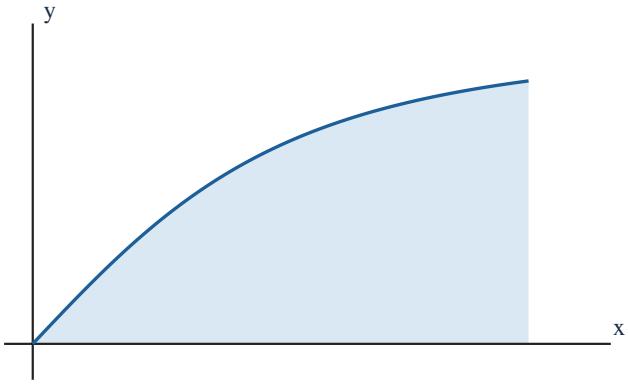


Change variable

Evaluate

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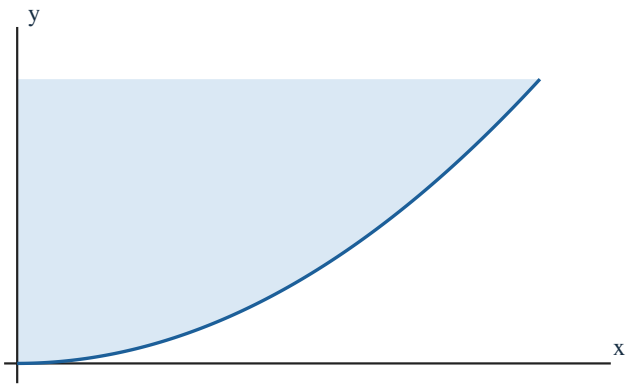
A curve has $x = \tan\theta,\ y = \sin\theta$. Rotate the region under it, $0 \leq x \leq \sqrt{3}$, about the x-axis. Find the exact volume.



Limits

Volume

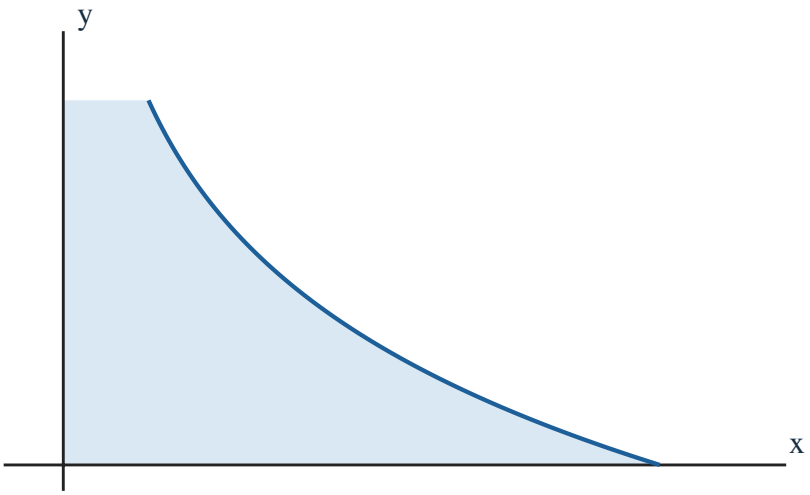
6. The curve $x = 2t,\ y = t^2, -2 \leq t \leq 2$, meets the horizontal line $y = a$ at its endpoints. Find a and the volume of the bounded region rotated about the y-axis.



Set up

Volume

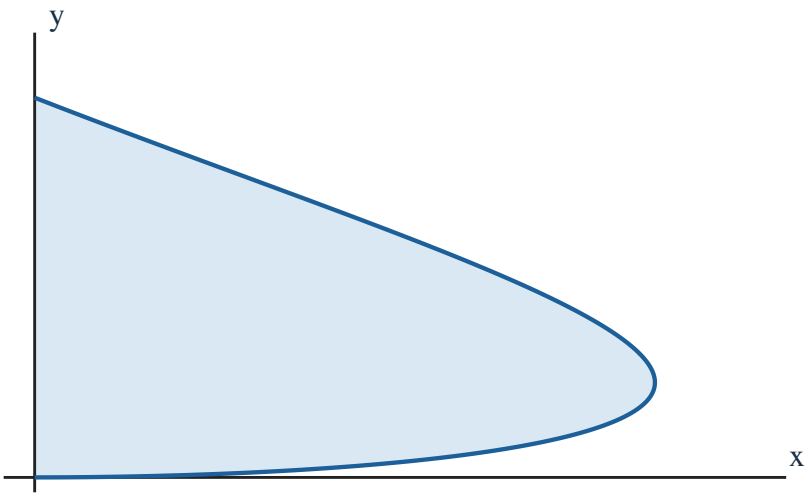
8. A curve has $x = 1/(2t),\ y = \ln(2t),\ t \geq 1/2$. The region between it, the coordinate axes and $y = a$ rotates about the y-axis. Its volume is $24\pi/49$. Find a.



Integral

Solve

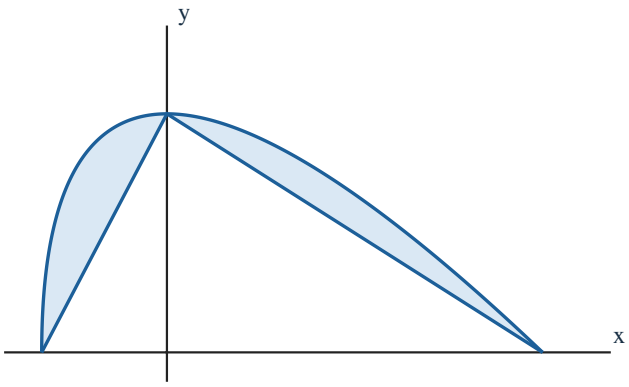
9. The curve $x = 2\sin t,\ y = t^2, 0 \leq t \leq \pi$, encloses a region with the y-axis. Find the volume when it rotates about that axis.



Set up

Integrate

10. The curve $x = t^2 - 2t,\ y = 1 - t^2,$ $-1 \leq t \leq 1$, intersects the axes at P, Q, S. Rotate the region between the curve and chords PQ, QS about the x-axis. Find its volume.



Intercepts

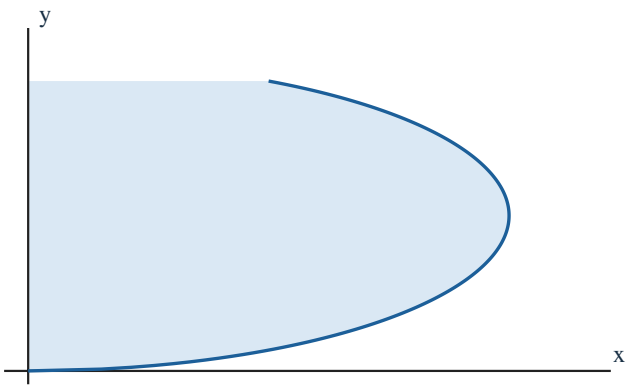
Subtract cones

Modelling with Volumes of Revolution 24

A goldfish bowl is modelled by rotating $x = 2\sin t,\ y = 2\cos t + 2, \pi/6 \leq t \leq 11\pi/6,$ about the y-axis.

a. Find the volume up to height 3 cm.

b. A similar real bowl has maximum diameter 48 cm. Find the corresponding water volume.



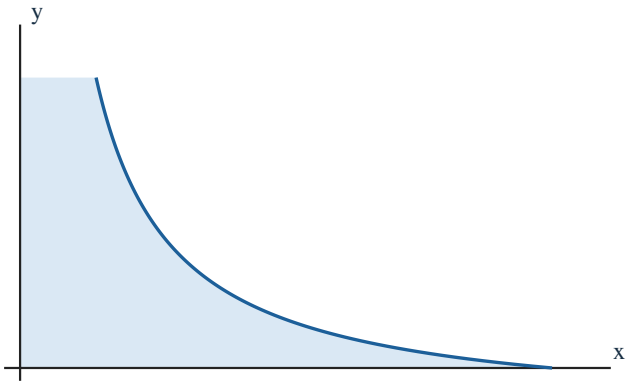
a

b

1. A volcano is modelled by rotating the region $0 \le x \le 2000/(20+y)$, $0 \le y \le 120$, about the y-axis. Units are metres.

a. Find the base diameter.

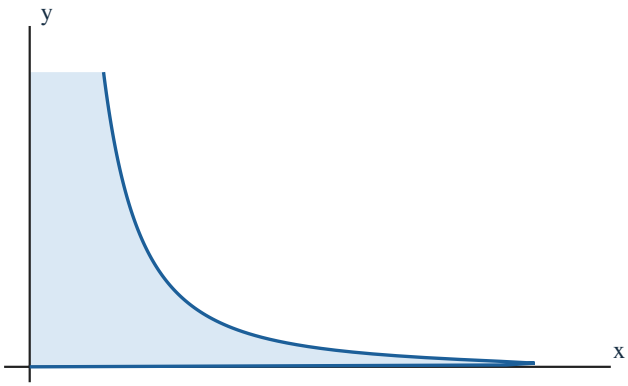
b. Estimate its volume.



a

b

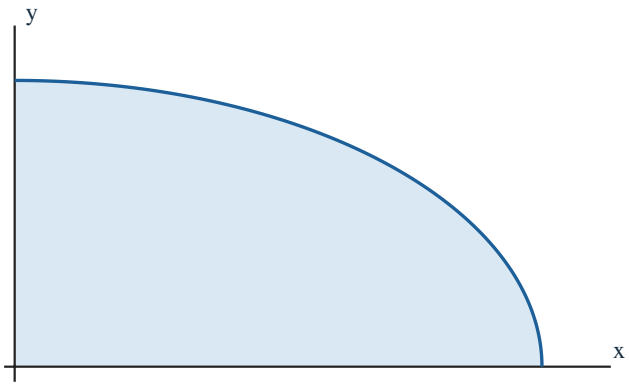
2. A vase of height 30 cm is formed by rotating $x^2 = 100y/(10y^2+1)$, $0 \le y \le 30$, about the y-axis. It is filled to height 20 cm. Find the exact water volume as $p\pi \ln q$ cm^3, for integers p, q.



2

3a. Prove $\cos^3\theta = \frac{3}{4}\cos\theta + \frac{1}{4}\cos 3\theta$.

b. A domed tent is formed by rotating $x = 50\cos\theta,\ y = 30\sin\theta, 0 \le \theta \le \pi/2$, about the y-axis. Find its volume in cubic metres.



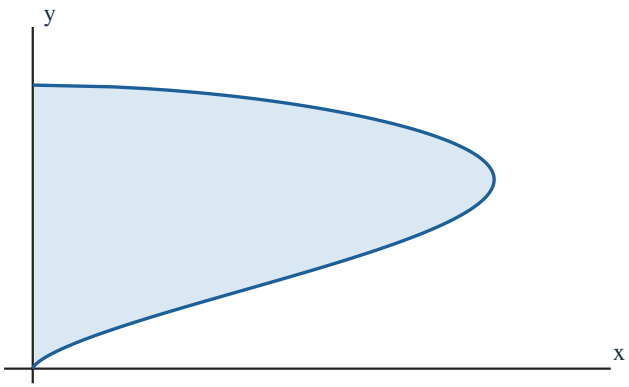
a

b

4. A balloon model is formed by rotating $x = \sin y\sqrt{\sin 2y}, 0 \le y \le \pi/2$, about the y-axis. Units are metres.

a. Find its volume.

b. The real balloon has height 6π metres. Find its volume.



a

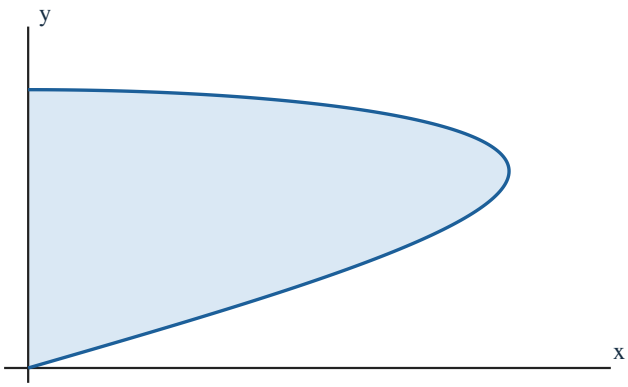
b

5. A silver earring is formed by rotating $x = 2\sin 2\theta,\ y = 3\sin\theta, 0 \leq \theta \leq \pi$, about the y-axis. Units are millimetres.

a. Show $x^2 = \frac{16}{81}y^2(9 - y^2)$.

b. Find the maximum number made from $300\ \text{mm}^3$ of silver.

c. Discuss modelling limitations.



a

b

c